CASES

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SOURCE / FIRM:	BOOZ ALLEN HAMILTON
DATE / CONTEXT:	2nd round interview for summer internship 2004
ISSUES COVERED:	Brainteaser ☐ Estimation ☒ Business Case ☐
ISSUE/PROBLEM POSED:	How would you value a football (soccer) player?
INFORMATION PROVIDED:	None
SOLUTION:	I went around the houses a bit on this one. Identified sources of revenue for a football club including: Ticket revenue, Revenue from TV coverage, Sponsorship Merchandise sales (e.g. shirts) Talked around how to determine the contribution that an individual player makes to those revenue streams The final conclusion was that the best way to divide the portion of revenue related to the actual players (over their lifetime) between the team would be based on individual ratings (like those published for each player in the fantasy football league) Talked around valuation as the present value of future cash flows related to each player
OTHER USEFUL TIPS:	Open discussion with structure maintained through making notes and drawing tree diagrams was appreciated – used diagrams as prompt for directing discussions

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SOURCE / FIRM:	ROLAND BERGER
DATE / CONTEXT:	2nd round interview for summer internship 2004
ISSUES COVERED:	Brainteaser ☐ Estimation ☒ Business Case ☐
ISSUE/PROBLEM POSED:	What is the size of the scrap metal market in Russia?
INFORMATION PROVIDED:	The reliable data to estimate the scrap metal market directly does not exist. The market is dominated by 4 underground players that do not reveal any data.
SOLUTION:	Solution: the data about the iron ore production and total output of still mills could be found. We can approximate how many tons of metal is possible to get from the iron ore. Subtracting potential production from actual production we can roughly estimate how many tons are coming from the scrap market.
OTHER USEFUL TIPS:	Estimation case that tests the ability to find the needed data from limited information provided.

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SOURCE / FIRM:	DELOITTE CONSULTING
DATE / CONTEXT:	1st round interview for Deloitte Internship 2002
ISSUES COVERED:	Brainteaser ☐ Estimation ☒ Business Case ☐
ISSUE POSED:	Determine the size of the Dutch market for teacups and plates from Brazil.
INFORMATION PROVIDED:	This information is given only if the candidate asks for it. Population of Netherlands: 16 million Average household: 1.8 person (rounded to 2) Average cups and plates per household: 6 sets Average time of replacement: 5 years. Other information given: read the solution
SOLUTION:	After the candidate had asked for the additional information above, he was asked to determine all possible sources of information for these data. In order to do so, he performed a supply chain analysis. He established sources of information between all nodes of supply chain and showed how to convert information obtained between nodes to final market size. He also analysed expected accuracy for each data source. The interviewer was not satisfied until all nodes were addressed: he kept asking which other sources could be found until the candidate drew a diagram of the supply chain and proved that all nodes had been addressed. Note that information can come from the supply chain has two starting points, Brazil and the Netherlands, and joins before distribution. All nodes in both branches must be addressed). After working on the information sources, the candidate was asked to go back to the original problem and solve it. With the information given, he found the market size for sets bought by households. The interviewer then asked him to list all other possible markets. From the list, he was asked to perform an analysis and determine the size of the market for hotels. He was told that no information on the hotel industry was available, but that general statistics on tourism could be asked. The candidate asked: • The amount of tourists coming to the Netherlands each year (20 million) • The average duration of their stay (7 days) => He calculated the average number of tourists in the Netherlands at any time • The average percentage of rooms occupied: 60% => He calculated number of hotel rooms in the Netherlands • he assumed 3 sets per room (which the interviewer accepted) • He assumed a faster replacement in hotels than in private households (he proposed 1 year but the interviewer told him that 2 years was more credible)
OTHER USEFUL TIPS:	

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SOURCE / FIRM:	**ROLAND BERGER**
DATE / CONTEXT:	1st round interview for Roland Berger Internship 2002
ISSUES COVERED:	Brainteaser ☐ Estimation ☒ Business Case ☐
ISSUE POSED:	You have to decide within 10 minutes whether or not to order 10 million ping-pong balls, for delivery the next day. Can you fit them all in one 747? If yes, do the deal, if not, don't.
INFORMATION PROVIDED:	
SOLUTION:	This is a classic guesstimate. You have to estimate the volume of 10 million ping-pong balls and the volume of a 747. You can for example approximate the volume of one ping-pong ball by the volume of a cube with a 3 cm side => volume of 10 millions ping-pong balls is 270 million cm3 = 270 m3 (remember that 1 m3 = 1,000,000 cm 3). To estimate the volume of a 747, you can use different approaches. You may happen to have a pretty good idea of the length and width of a 747 because you are flying in them all the time and simply use the cylinder volume formula (π*r2*height), correcting for non-usable space like the cockpit, the motors… You may remember how many people a 747 seats (approx. 500) and deduce the volume from there.
OTHER USEFUL TIPS:	

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SOURCE / FIRM:	**ROLAND BERGER**
DATE / CONTEXT:	1st round interview for Roland Berger Internship 2002
ISSUES COVERED:	Brainteaser ☐ Estimation ☒ Business Case ☐
ISSUE POSED:	How would you estimate the size of the market for aircraft maintenance service? Can you estimate its growth? (The timeline for the estimate is one or two days.)
INFORMATION PROVIDED:	On questioning the interviewer narrowed the area to commercial aircraft and revealed that 80% was done in-house by commercial airlines. The other 20% was sub-contracted, and after further probing he revealed that there was one listed company among the contractors.
SOLUTION:	The point here is to show that you would know where to look to gather information. By examining the expenses and revenues of the listed company, as well as the type of aircraft they serviced, one could extrapolate figures to the world fleet as published by Boeing and Airbus. For growth, the two key drivers are numbers of each type of aircraft and their age, each of which could be gotten form the published fleet numbers and the projections thereof.
OTHER USEFUL TIPS:	

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SOURCE / FIRM:	**UNKNOWN**
DATE / CONTEXT:	LBS Interview – 1996
ISSUES COVERED:	Brainteaser ☒ Estimation ☐ Business Case ☐
ISSUE POSED:	How many gas stations are there in the US?
INFORMATION PROVIDED:	None
SOLUTION:	There are two ways I thought of to approach this question: 1. **Population theory** - In say Menlo Park / Atherton, there are about 15 gas stations for about 40,000 people. In urban areas, there are many more people per gas station (say 150 in San Francisco for 700,000 people), while in rural areas, there are fewer people per station (in my hometown, there were 3 gas stations for about 5,000 people). Add in a fudge factor for truck stops in the middle of nowhere, and let's guestimate that the average nation-wide (I know one isn't supposed to average averages, but this is consulting) is around as populated with stations as Menlo park. This is a ratio of 1 station per 2,667 people. There are 250 million people in the US so that's around 90,000 stations. Note - one could just have used Menlo Park from the start, but the idea is not to get the answer but to think transparently. I threw in things like urban areas, people without cars, commercial transport etc to show that I was casting a net to test the reasonableness of my assumptions, regardless of the fact that I finished where I started. 2. **Patterns of demand and a little knowledge** - My idea of the average gas station has 8 pumps. I have observed that, on average, 4 pumps are in use during the 14 hours a day the station (average station) is open. Let's guess that the average station sells (14 hours x 6 fills / hr x 4 pumps x 10 gallons of gas). I.e. 3,360 gallons of gas / day. That's around 1.2 million gallons per year. Now, I know that all of the US could fit into the front seats of all of the cars in the US, so let's assume there are 125 million cars on the road. If each car is driven for 12,000 miles at 20 miles per gallon, that implies (125 million x 12,000 miles/20mpg) i.e. 75 billion gallons of gas are consumed each year. Therefore 75 billion gallons / 1.2 million gallons / station / year = 62,500 stations. Note - I know that this is convoluted but more elegant solutions are available. Last I heard the right answer is around 80,000.
OTHER USEFUL TIPS:	Remember the idea is not to get the right answer but to think logically, in a linear fashion, and get to a reasonable answer. That is 1 million stations is obviously too many, where 1,000 is probably the number you personally have been to in your life. Given the interview conditions, it is worthwhile picking relatively easy numbers to work with rather than getting caught up in some highly complex mental arithmetic.

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SOURCE / FIRM:	**UNKNOWN**
DATE / CONTEXT:	LBS Interview – 1996
ISSUES COVERED:	Brainteaser ☐ Estimation ☒ Business Case ☐
ISSUE POSED:	You are in a meeting with a client who mentions that she is considering building a new plant. The new plant will require 100 million tons per year of recycled aluminium as an input. Your client turns to you and asks you if there is 100 million tons of recycled aluminium available in the US on a yearly basis. You do not have that information of the top of your heard. How can you answer the question on the spot?
INFORMATION PROVIDED:	None
SOLUTION:	• I know that soda cans are made of aluminium. Let's assume that soda cans are the major source of recycled aluminium. Also, let's assume that people drink 5 cans of soda per day. • 350 days / year x 5 cans / day / person = 1,750 cans per year per person • Let's assume there are 17.5 cans in a pound of aluminium. • That means there are 100 pounds of aluminium per year per person • There are 250 million people in the United States • That means that there are 25,000 million pounds per year. • Since 2,000 pounds = 1 ton, there are only 12.5 million tons of recycled aluminium available per year. Thus there is not enough recycled aluminium available per year in the United States.
OTHER USEFUL TIPS:	**WHAT NOT TO DO** • Make the math too difficult for yourself. It is acceptable and very wise to round off. For example I used 17.5 cans in a pond and 350 days in a year as they are close enough and they make the calculations easier. • Forget to state your assumptions - there are several assumptions you'll have to make to come to an answer. Make sure you state what they are. It is better to make an assumption that you are uncertain of rather than to stop and not to get to an answer. Once you have an answer, it is perfectly acceptable, and advisable to say "I've made several assumptions to come to this answer. One I am not sure of is my assumption about how many cans make up a pound. I said 17.4 cans are in a pound. If there were really twice that many, I would have to adjust my number as accordingly. Of course that would not change my bottom line answer. There would still not be enough recycled aluminium.

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Interview Questions: Finance Interview Brainteasers by Vault

Perhaps even more so than tough finance questions, brainteasers can unnerve the most icy-veined, well-prepared finance candidate. Even if you know the relationships between inflation, bond prices and interest rates like the back of a dollar bill, all your studying may not help you when your interviewer asks you how many ping pong balls fit in a 747.

That is partly their purpose. Investment bankers and other finance professionals need to be able to work well under pressure, so many interviewers believe that throwing a brainteaser at a candidate is a good way to test an applicant's battle-worthiness. But these questions serve another purpose, too - interviewers want you to showcase your ability to analyze a situation, and to form conclusions about this situation. It is not usually important that you come up with a "correct" answer, just that you display strong analytical ability.

Remember, brainteasers are very unstructured, so it is tough to suggest a step-by-step methodology. There are a couple of set rules, though. First, take notes as your interviewer gives you a brainteaser, especially if it's heavy on the math. Second, think aloud so your interviewer can hear your thought process. Here are some samples:

1. *If you look at a clock and the time is 3:15, what is the angle between the hour and the minute hands?*

The answer to this is not zero! The hour hand, remember, moves as well. The hour hand moves a quarter of the way between three and four, so it moves a quarter of a twelfth (1/48) of 360 degrees. So the answer is seven and a half degrees, to be exact.

2. *You have a five-gallon jug and a three-gallon jug. You must obtain exactly four gallons of water. How will you do it?*

You should find this brainteaser fairly simple. If you were to think out loud, you might begin by examining the ways in which combinations of five and three can come up to be four. For example: (5 - 3) + (5 - 3) = 4. This path does not actually lead to the right answer, but it is a fruitful way to begin thinking about the question. Here's the solution: fill the three-gallon jug with water and pour it into the five-gallon jug. Repeat. Because you can only put two more gallons into the five-gallon jug, one gallon will be left over in the three-gallon jug. Empty out the five-gallon jug and pour in the one gallon. Now just fill the three-gallon jug again and pour it into the five-gallon jug. Ta-da. (Mathematically, this can be represented 3 + 3 - 5 + 3 = 4)

3. *You are faced with two doors. One door leads to your job offer (that's the one you want!), and the other leads to the exit. In front of each door is a guard. One guard always tells the truth. The other always lies. You can ask one question to decide which door is the correct one. What will you ask?*

The way to logically attack this question is to ask how you can construct a question that provides the same answer (either a true statement or a lie), no matter who you ask.

There are two simple answers. Ask a guard: "If I were to ask you if this door were the correct one, what would you say?" The truthful consultant would answer yes (if it's the correct one), or no (if it's not). Now take the lying consultant. If you asked the liar if the correct door is the

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right way, he would answer no. But if you ask him: "If I were to ask you if this door were the correct one, what would you say," he would be forced to lie about how he would answer, and say yes. Alternately, ask a guard: "If I were to ask the other guard which way is correct, what would he say?" Here, the truthful guard would tell you the wrong way (because he is truthfully reporting what the liar would say), while the lying guard would also tell you the wrong way (because he is lying about what the truthful guard would say).
If you want to think of this question more mathematically, think of lying as represented by -1, and telling the truth as represented by +1. The first solution provides you with a consistently truthful answer because (-1)(-1) = 1, while (1)(1) = 1. The second solution provides you with a consistently false answer because (1)(-1) = -1, and (-1)(1) = -1.

VAULT

1. How many gallons of white house paint are sold in the U.S. every year?

THE "START BIG" APPROACH: If you're not sure where to begin, start with the basic assumption that there are 270 million people in the U.S. (or 25 million businesses, depending on the question). If there are 270 million people in the United States, perhaps half of them live in houses (or 135 million people). The average family size is about three people, so there would be 45 million houses in the United States. Let's add another 10 percent to that for second houses and houses used for other purposes besides residential. So there are about 50 million houses.

If houses are painted every 10 years, on average (notice how we deftly make that number easy to work with), then there are 5 million houses painted every year. Assuming that one gallon of paint covers 100 square feet of wall, and that the average house has 2,000 square feet of wall to cover, then each house needs 20 gallons of paint. So 100 million gallons of paint are sold per year (5 million houses x 20 gallons). (Note: If you want to be fancy, you can ask your interviewer whether you should include inner walls as well!) If 80 percent of all houses are white, then 80 million gallons of white house paint are sold each year. (Don't forget that last step!)

(more of this guesstimate on the next page)

THE "START SMALL" APPROACH: You could also start small, and take a town of 27,000 (about 1/10,000 of the population). If you use the same assumption that half the town lives in houses in groups of three, then there are 4,500 houses, plus another 10 percent, then there are really 5,000 houses to worry about. Painted every 10 years, 500 houses are being painted in any given year. If each house has 2,000 square feet of wall, and each gallon of paint covers 100 square feet, then each house needs 20 gallons - and so 10,000 gallons of house paint are sold each year in your typical town. Perhaps 8,000 of those are white. Multiply by 10,000 - you have 80 million gallons.

Your interviewer may then ask you how you would actually get that number, on the job, if necessary. Use your creativity - contacting major paint producers would be smart, putting in a call to HUD's statistics arm could help, or even conducting a small sample of the second calculation in a few representative towns is possible.

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2. What is the size of the market for disposable diapers in China?

Here's a good example of a market sizing. How many people live in China? A billion. Because the population of China is young, a full 600 million of those inhabitants might be of child-bearing age. Half are women, so there are about 300 million Chinese women of childbearing age. Now, the average family size in China is restricted, so it might be 1.5 children, on average, per family. Let's say two-thirds of Chinese women have children. That means that there are about 200 million children in China. How many of those kids are under the age of two? About a tenth, or 20 million. So there are at least 20 million possible consumers of disposable diapers.

To summarize:

1 billion people x 60% childbearing age = 600,000,000 people
600,000,000 people x 1/2 are women = 300,000,000 women of childbearing age
300,000,000 women x 2/3 have children = 200,000,000 women with children
200,000,000 women x 1.5 children each = 300,000,000 children
300,000,000 children x 1/10 under age 2 = 30 million

3. How many square feet of pizza are eaten in the United States each month?

Take your figure of 300 million people in America. How many people eat pizza? Let's say 200 million. Now let's say the average pizza-eating person eats pizza twice a month, and eats two slices at a time. That's four slices a month. If the average slice of pizza is perhaps six inches at the base and 10 inches long, then the slice is 30 square inches of pizza. So four pizza slices would be 120 square inches. Therefore, there are a billion square feet of pizza eaten every month.

To summarize:

300 million people in America
200 million eat pizza
Average slice of pizza is six inches at the base and 10 inches long = 30 square inches (height x half the base)
Average American eats four slices of pizza a month
Four pieces x 30 square inches = 120 square inches (one square foot is 144 inches), so let's assume one square foot per person
200 million square feet a month

4. How would you estimate the weight of the Chrysler building?

This is a process guesstimate - the interviewer wants to know if you know what questions to ask. First, you would find out the dimensions of the building (height, weight, depth). This will allow you to determine the volume of the building. Does it taper at the top? (Yes.) Then, you need to estimate the composition of the Chrysler building. Is it mostly steel? Concrete? How much would those components weigh per square inch? Remember the extra step - find out whether you're considering the building totally empty or with office furniture, people, etc.? (If you're including the contents, you might have to add 20 percent or so to the building's weight.)

5. Why are manhole covers round?

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The classic brainteaser, straight to you via Microsoft (the originator). Even though this question has been around for years, interviewees still encounter it.

Here's how to "solve" this brainteaser. Remember to speak and reason out loud while solving this brainteaser!

Why are manhole covers round? Could there be a structural reason? Why aren't manhole covers square? It would make it harder to fit with a cover. You'd have to rotate it exactly the right way. So many manhole covers are round because they don't need to be rotated. There are no corners to deal with. Also, a round manhole cover won't fall into a hole because it was rotated the wrong way, so it's safer.

Looking at this, it seems corners are a problem. You can't cut yourself on a round manhole cover. And because it's round, it can be more easily transported. One person can roll it.

6. If you look at a clock and the time is 3:15, what is the angle between the hourand the minute hands?

The answer to this is not zero! The hour hand, remember, moves as well. The hour hand moves a quarter of the way between three and four, so it moves a quarter of a twelfth (1/48) of 360 degrees. So the answer is seven and a half degrees, to be exact.

7. You have a five-gallon jug and a three-gallon jug. You must obtain exactly four gallons of water. How will you do it?

You should find this brainteaser fairly simple. If you were to think out loud, you might begin by examining the ways in which combinations of five and three can come up to be four. For example: (5 - 3) + (5 - 3) = 4. This path does not actually lead to the right answer, but it is a fruitful way to begin thinking about the question. Here's the solution: fill the three-gallon jug with water and pour it into the five-gallon jug. Repeat. Because you can only put two more gallons into the five-gallon jug, one gallon will be left over in the three-gallon jug. Empty out the five-gallon jug and pour in the one gallon. Now just fill the three-gallon jug again and pour it into the five-gallon jug. Ta-da. (Mathematically, this can be represented 3 + 3 - 5 + 3 = 4)

8. You have 12 balls. All of them are identical except one, which is either heavier or lighter than the rest. The odd ball is either hollow while the rest are solid, or solid while the rest are hollow. You have a scale, and are permitted three weighings. Can you identify the odd ball, and determine whether it is hollow or solid?

This is a pretty complex question, and there are actually multiple solutions. First, we'll examine what thought processes an interviewer is looking for, and then we'll discuss one solution.

Start with the simplest of observations. The number of balls you weigh against each other must be equal. Yeah, it's obvious, but why? Because if you weigh, say three balls against five, you are not receiving any information. In a problem like this, you are trying to receive as much information as possible with each weighing.

For example, one of the first mistakes people make when examining this problem is that they believe the first weighing should involve all of the balls (six against six). This weighing involves all of the balls, but what type of information does this give you? It actually gives you

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no new information. You already know that one of the sides will be heavier than the other, and by weighing six against six, you will simply confirm this knowledge. Still, you want to gain information about as many balls as possible (so weighing one against one is obviously not a good idea). Thus the best first weighing is four against four.

Secondly, if you think through this problem long enough, you will realize how precious the information gained from a weighing is: You need to transfer virtually every piece of information you have gained from one weighing to the next. Say you weigh four against four, and the scale balances. Lucky you! Now you know that the odd ball is one of the unweighed four. But don't give into the impulse to simply work with those balls. In this weighing, you've also learned that the eight balls on the scale are normal. Try to use this information.

Finally, remember to use your creativity. Most people who work through this problem consider only weighing a number of balls against each other, and then taking another set and weighing them, etc. This won't do. There are a number of other types of moves you can make - you can rotate the balls from one scale to another, you can switch the balls, etc.

Let's look at one solution:

(more of this brainteaser on next page)

For simplicity's sake, we will refer to one side of the scale as Side A, and the other as Side B.

Step 1: Weigh four balls against four others.

Case A: If, on the first weighing, the balls balance
If the balls in our first weighing balance we know the odd ball is one of those not weighed, but we don't know whether it is heavy or light. How can we gain this information easily? We can weigh them against the balls we know to be normal. So:

> **Step 2 (for Case A):** Put three of the unweighed balls on the Side A; put three balls that are known to be normal on Side B.
>
> > I. If on this second weighing, the scale balances again, we know that the final unweighed ball is the odd one.
>
> **Step 3a (for Case A):** Weigh the final unweighed ball (the odd one) against one of the normal balls. With this weighing, we determine whether the odd ball is heavy or light.
>
> > II. If on this second weighing, the scale tips to Side A, we know that the odd ball is heavy. (If it tips to Side B, we know the odd ball is light, but let's proceed with the assumption that the odd ball is heavy.) We also know that the odd ball is one of the group of three on Side A.
>
> **Step 3b (for Case A):** Weigh one of the balls from the group of three against another one. If the scale balances, the ball from the group of three that was unweighed is the odd ball, and is heavy. If the scale tilts, we can identify the odd ball, because we know it is heavier than the other. (If the scale had tipped to Side B, we would use the same logical process, using the knowledge that the odd ball is light.)

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Case B: If the balls do not balance on the first weighing

If the balls do not balance on the first weighing, we know that the odd ball is one of the eight balls that was weighed. We also know that the group of four unweighed balls are normal, and that one of the sides, let's say Side A, is heavier than the other (although we don't know whether the odd ball is heavy or light).

Step 2 (for Case B): Take three balls from the unweighed group and use them to replace three balls on Side A (the heavy side). Take the three balls from Side A and use them to replace three balls on Side B (which are removed from the scale).

I. If the scale balances, we know that one of the balls removed from the scale was the odd one. In this case, we know that the ball is also light. We can proceed with the third weighing as described in step 3b from Case A.

II. If the scale tilts to the other side, so that Side B is now the heavy side, we know that one of the three balls moved from Side A to Side B is the odd ball, and that it is heavy. We proceed with the third weighing as described in step 3b in Case A.

III. If the scale remains the same, we know that one of the two balls on the scale that was not shifted in our second weighing is the odd ball. We also know that the unmoved ball from Side A is heavier than the unmoved ball on Side B (though we don't know whether the odd ball is heavy or light).

Step 3 (for Case B): Weigh the ball from Side A against a normal ball. If the scale balances, the ball from Side B is the odd one, and is light. If the scale does not balance, the ball from Side A is the odd one, and is heavy.

(more of this brainteaser on next page)

Whew! As you can see from this solution, one of the keys to this problem is understanding that information can be gained about balls even if they are not being weighed. For example, if we know that one of the balls of two groups that are being weighed is the odd ball, we know that the unweighed balls are normal. Once this is known, we realize that breaking the balls up into smaller and smaller groups of three (usually eventually down to three balls), is a good strategy - and an ultimately successful one.

9. You are faced with two doors. One door leads to your job offer (that's the one you want!), and the other leads to the exit. In front of each door is a guard. One guard always tells the truth. The other always lies. You can ask one question to decide which door is the correct one. What will you ask?

The way to logically attack this question is to ask how you can construct a question that provides the same answer (either a true statement or a lie), no matter who you ask.

There are two simple answers. Ask a guard: "If I were to ask you if this door were the correct one, what would you say?" The truthful consultant would answer yes (if it's the correct one), or no (if it's not). Now take the lying consultant. If you asked the liar if the correct door is the right way, he would answer no. But if you ask him: "If I were to ask you if this door were the correct one, what would you say," he would be forced to lie about how he would answer, and say yes. Alternately, ask a guard: "If I were to ask the other guard which way is correct, what would he say?" Here, the truthful guard would tell you the wrong way (because he is

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truthfully reporting what the liar would say), while the lying guard would also tell you the wrong way (because he is lying about what the truthful guard would say).

If you want to think of this question more mathematically, think of lying as represented by -1, and telling the truth as represented by +1. The first solution provides you with a consistently truthful answer because (-1)(-1) = 1, while (1)(1) = 1. The second solution provides you with a consistently false answer because (1)(-1) = -1, and (-1)(1) = -1.

10. A company has 10 machines that produce gold coins. One of the machines is producing coins that are a gram light. How do you tell which machine is making the defective coins with only one weighing?

Think this through - clearly, every machine will have to produce a sample coin or coins, and you must weigh all these coins together. How can you somehow indicate which coins came from which machine? The best way to do it is to have every machine crank a different number of coins, so that machine 1 will make one coin, machine 2 will make two coins, and so on. Take all the coins, weigh them together, and consider their weight against the total theoretical weight. If you're four grams short, for example, you'll know that machine 4 is defective.

11. The four members of U2 (Bono, the Edge, Larry and Adam) need to get across a narrow bridge to play a concert. Since it's dark, a flashlight is required to cross, but the band has only one flashlight, and only two people can cross the bridge at a time. (This is not to say, of course, that if one of the members of the band has crossed the bridge, he can't come back by himself with the flashlight.) Adam takes only a minute to get across, Larry takes two minutes, the Edge takes five minutes, and slowpoke Bono takes 10 minutes. A pair can only go as fast as the slowest member. They have 17 minutes to get across. How should they do it?

The key to attacking this question is to understand that Bono and the Edge are major liabilities and must be grouped together. In other words, if you sent them across separately, you'd already be using 15 minutes. This won't do. What does this mean? That Bono and the Edge must go across together. But they can not be the first pair (or one of them will have to transport the flashlight back).

Instead, you send Larry and Adam over first, taking two minutes. Adam comes back, taking another minute, for a total of three minutes. Bono and the Edge then go over, taking 10 minutes, and bringing the total to 13. Larry comes back, taking another two minutes, for a total of 15. Adam and Larry go back over, bringing the total time to 17 minutes.

12. What is the decimal equivalent of 3/16 and 7/16?

A commonly-used Wall Street interview question, this one isn't just an attempt to stress you out or see how quick your mind works. This question also has practical banking applications. Stocks often are traded at prices reported in 1/16s of a dollar. (Each 1/16 = .0625, so 3/16 = .1875 and 7/16 = .4375).

13. What is the sum of the numbers from one to 50?

Another question that recent analyst hires often report receiving. This is a relatively easy one: pair up the numbers into groups of 51 (1 + 50 = 51; 2 + 49 = 51; etc.). Twenty-five pairs of 51 equals 1275.

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14. You have a painting that is $320 that is selling for 20 percent off. How much is the discounted price?

Calculate quickly: What's 80 percent of $320? The answer's $256. Even in a question like this, if you are good with numbers and use shortcuts, don't be afraid to talk aloud. For example: 80 percent of $320 can be broken down to a calculation like 80 percent of $80 x $4, or 162.

15. You're playing three-card monte. Two cards are red, one is black. (Note: In three-card monte, the three cards are face down and you try to pick the black card in order to win.) You pick the middle card. After you pick, the dealer shows that one of the cards you have not chosen is red. You are given the chance to switch your selection. Should you?

The short answer is yes. By switching, you are betting that the card you initially chose was red. By not switching, you are betting that the card you initially chose was black. And because two out of three cards are red, of course, betting on red is the way to go.

Let's break it down, starting with the not switching case. Say the first card you chose was the black one. This happens one-third of the time. If you do not switch your choice, you win. Needless to say, the other two-thirds of the time, having picked a red card, and deciding not to switch, you lose. In other words, if you do not switch, you win a third of the time.

Now let's examine what happens when you switch cards. Say the first card you chose was the black one. Again, this would happen one-third of the time. If, after being shown a red card, you switch, you lose. The other two-thirds of the time, if you switch, you win because the dealer has already shown you that one of the cards you did not pick is red. Given the premise that your original pick was a red card, the card you are switching to must be the black one. You will win two-thirds of the time.

16. A straight flush beats a four-of-a-kind in poker because it is more unlikely. But think about how many straight flushes there are - if you don't count wraparound straights, you can have a straight flush starting on any card from two to 10 in any suit (nine per suit). That means there are 36 straight flushes possible. But how many four of a kinds are there - only 13. What's wrong with this reasoning?

Immediately, you should think about what the difference is between a straight flush and a four-of-a-kind. One involves five cards, and the other involves four. Intuitively, that's what should strike you as the problem with the line of reasoning. Look closer and you'll see what that means: for every four of a kind, there are actually a whole bunch of five-card hands: 48 (52 - 4) in fact. There are actually 624 (48 x 13) of them in all.

17. If you have seven white socks and nine black socks in a drawer, how many do you have to pull out blindly in order to ensure that you have a matching pair?

Three. Let's see - if the first one is one color, and the second one is the other color, the third one, no matter what the color, will make a matching pair. Sometimes you're not supposed to think that hard.

18. Tell me a good joke that is neither sexist nor racist.

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If you can't think of any, you're in the same boat as the unfortunately tongue-tied recent candidate at Salomon Smith Barney. Find one and remember it.

19. If I were to fill this room with pennies, how many pennies would fit in?

A literally in-your-face guesstimate.

20. Say you are driving on a one-mile track. You do one lap at 30 miles an hour. How fast do you have to go to average 60 miles an hour?

This is something of a trick question, and was recently received by a Goldman candidate. The first thought of many people is to say 90 miles an hour, but consider: If you have done a lap at 30 miles an hour, you have already taken two minutes. Two minutes is the total amount of time you would have to take in order to average 60 miles an hour. Therefore, you can not average 60 miles an hour over the two laps.

*Guesstimates are commonly asked in consulting and investment banking interviews. Generally, your interviewer asks you to estimate the number or size of something, and observes your reasoning process. Most interviewers don't care if you actually get the correct number - what they want to see is that you are able to logically think through a process, creatively think through any possible exceptions or short cuts, and calculate basic sums in your head. You won't be given any real data (though you won't need to know much more beyond the fact that the United States has about 270 million inhabitants and 25 million businesses), and you shouldn't request any; it's irrelevant to the problem at hand. Make reasonable assumptions, with easy-to-work-with numbers, and go from there (remember that you're expected to use a pen and notepad to work through your calculations). These guesstimates may also involve elements of creativity and problem solving. For example, when posed the question "How much change would you find on the floor of a mall?" you might want to ask "Is there a fountain in the mall?"

Sample guesstimate:

1. How many gallons of white housepaint are sold in the U.S. each year?

THE "START BIG" APPROACH: If you're not sure where to begin, start with the basic assumption that there are 270 million people in the U.S. (or 25 million businesses, depending on the question). If there are 270 million people in the United States, perhaps half of them live in houses (or 135 million people). The average family size is about 3, so there would be 45 million houses in the United States. Let's add another 10 percent to that for second houses and houses used for other purposes besides residential. So there are about 50 million houses.

If houses are painted every 10 years on average (notice how we deftly make that number easy to work with), then there are 5 million houses painted every year. Assuming that one gallon of paint covers 100 square feet of wall, and that the average house has 2000 square feet of wall to cover, then each house needs 20 gallons of paint. So 100 million gallons of paint are sold per year (5 million houses x 20 gallons). (Note: If you want to be fancy, you can ask your interviewer whether you should include inner walls as well!) If 80 percent of all houses are white, then 80 million gallons of white housepaint are sold each year. (Don't forget that last step!)

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passengers? Are companies in that industry doing anything particularly well that may apply to the airline business?)

Chapter 7: **Guesstimates**

"Guesstimate" case interviews

Guesstimate questions are among the most unnerving questions you may ever have to answer in an interview situation. They can be so "off the wall" as to shake up an otherwise calm, collected candidate.

The approach to guesstimates is basically the same as business cases - you will showcase your ability to analyze a situation and form conclusions about this situation by thinking out loud. The difference here is that you will not necessarily be using a series of questions to gather feedback from the interviewer. Instead, you will drive toward a conclusion through a series of increasingly specific statements. Let's look at an example:

How many ping-pong balls fit in a 747?

No, this isn't a joke. This is an actual question used in consulting interviews. If you are a little unsettled by this type of question, it's no wonder. That is exactly the reaction the interviewer is expecting. Remember that the main objective of these questions is to evaluate your poise and professionalism when facing an outlandish situation. How you react to this question when presented will speak volumes about your ability to be professional when faced with a similar business situation at a client.

So, how do you approach a guesstimate question? First, do NOT panic. If you are visibly shaken when presented with a guesstimate or brainteaser, it will hurt you. It is extremely important that you do not lose your cool.

Do not let yourself struggle verbally. You are free to say something like, "That is an intriguing question. May I have a moment to think it through?" This statement immediately shows the interviewer you are still in control and gives you some breathing time to think about a method for answering.

Once you have had a minute to compose your thoughts, be sure and go through your reasoning out loud, so your interviewer can see that you're arriving at your answer in a logical manner.

"Don't be anal," suggests one former consultant. "You should realize that for the purposes of a guesstimate, 1,000,553 is the same as a million, and you can divide by 350 if you need to divide by the number of days in the year."

Finally, remember that there is no right answer for guesstimates. It will often not even be necessary to come up with a definitive response like "1,400,350," due to constraints on time. Always work toward a final answer, but do not feel that you have done a poor job if the interviewer moves on to other topics before you are finished. They may simply recognize that you're on the right track and see no reason to keep going.

Acing guesstimates

The best approach for a guesstimate or brainteaser question is to think of a funnel. You begin by thinking broadly, then slowly drill down towards the answer. Let's look at this approach in context.

Referring to our sample question, you know that you are looking for how many ping-pong balls fit in a 747 airplane. The first thing you need to determine is the volume of the ping-pong ball.
For any guesstimate or brainteaser question you will need to understand whether your interviewer will be providing any direction or whether you will have to make assumptions. Therefore, begin the analysis of a guesstimate or brainteaser question with a question to your interviewer, such as, "What is the volume of a single ping pong ball?" If the interviewer does not know or refuses to provide any answer, then you will know that you must assume the answer. If the interviewer does provide the information, then your approach will be a series of questions. For this example let's assume your interviewer wants you to make the assumptions. Your verbal dialogue might go something like this:
Let's assume that the volume of a ping-pong ball is three cubic inches. Now let's assume that all the seats in the plane are removed. We'll say the average person is six feet high, one foot wide and one foot deep. That's 6 cubic feet, or 10,368 cubic inches. (One cubic foot is 12x12x12 inches, or 1,728 cubic inches.)
Okay, so a 747 has about 400 seats in it, excluding the galleys, lavatories, and aisles on the lower deck and about 25 seats on the upper deck. Let's assume there are three galleys, 14 lavatories, and three aisles (two on the lower deck and one on the upper deck), and that the space occupied by the galleys is a six-person equivalent, by the lavatories is a two-person equivalent, and the aisles are a 50-person equivalent on the lower deck and a 20-person equivalent on the upper deck. That's an additional 18, 28, and 120 person-volumes for the remaining space. We won't include the cockpit since someone has to fly the plane. So there are about 600 person-equivalents available. (You would be rounding a bit to make your life easier, since the actual number is 591 person equivalents.)
In addition to the human volume, we have to take into account all the cargo and extra space - the belly holds, the overhead luggage compartments, and the space over the passengers' head. Let's assume the plane holds four times the amount of extra space as it does people, so that would mean extra space is 2,400 person-equivalents in volume. (Obviously, this assumption is the most important factor in this guesstimate. Remember that it's not important that this assumption be correct, just that you know the assumption should be made.)

Therefore, in total we have 3,000 (or 600 + 2,400) person-equivalents in volume available. Three thousand x 10,368 cubic inches means we have 31,104,000 cubic inches of space available. At three cubic inches per ball, a 747 could hold 10,368,000 balls. However, spheres do not fit perfectly together. Eliminate a certain percentage - spheres cover only about 70 percent of a cube when packed - and cut your answer to 7,257,600 balls.
You might be wondering how you would calculate all these numbers in your head! No one expects you to be a human calculator, so you should be writing down these numbers as you develop them. Then you can do the math on paper, in front of the interviewer, which will further demonstrate your analytical abilities.
You choose the numbers, so pick nice round numbers that are easy for you to manipulate. Even if you just read a study that states that there are 270 million inhabitants in the United States, no interviewer will flinch if you estimate the number of American inhabitants as 300 million.

Quick Guesstimate Numbers

You'll need to grab numbers for guesstimates quickly. Here are some basic stats.
There are approximately 260 million people in the United States (but you can round up to 300 million for the purposes of guesstimates). By comparison, there are 1.2 billion people in China.
There are about 100 million households in the United States.
There are 60 million people online in the United States.
According to Gemini Consulting, 89 percent of all Web users have English as a first language.

Estimation Cases

Suppose you are flying on a plane with the CEO from Exxon and you want to sell a consulting engagement. He has just left to use the lavatory and you have about five minutes to estimate his yearly revenues from personal automobile gasoline sales in the U.S. (excluding commercial trucks, boats, etc.) How would you go about coming up with this estimate?

1. Assume the population of the U.S. is 250 million.
2. Estimated number of people per household is 2.5, making 100 million households.
3. Estimated number of cars per household is 1, which gives 100 million cars in the U.S.
4. Assume each car gets filled up once per week (or 50 times per year to use simple numbers)
5. Assume the average fill-up is 10 gallons. 50 X 10 is 500 gallons per car.
6. Total gallons sold is 500 X 100 million = 50 billion.
7. If average price is $1.25, total revenue from U.S. automobiles is $62.5 billion.
8. Estimated market share of Exxon is 20% [the interviewer asked me why and I explained that I believed the market was basically an oligopoly with a few players dominating the market. This type of market typically has market share of the dominating competitors of around 20%].
9. Calculated total revenues for Exxon from the U.S. household automobile market, therefore, is $12.5 billion.

How many beer bottles are currently in circulation in the US?

First I decided to figure out the annual beer consumption to get at annual consumption of bottles. I estimated the population of U.S. as 250M, took out children who don't consume beer (approximately 10%, which gave me the number of 225M). Then I divided it into men and women as they have different consumption patterns – men probably consume more. I estimated the number of men and women to be approximately equal at 125M and 125M. Then I estimated that men probably drink 2 bottles a week on average, making it approximately 100 bottles a year per person (heavy beer drinkers and men not drinking beer will average out), giving a total of 12.5B. Women probably drink 2 bottles a month making it a total of 3B yearly. The total yearly consumption is 15.5B.

This is where the trick was because my interviewer was not satisfied with a yearly number, he wanted to have a current circulation number. I used the concept of velocity to come up with the number: yearly consumption = current circulation * some velocity (# times the bottle goes through the economy). I estimated the velocity to be around 70 assuming an average 5-day purchasing cycle. Thus the current circulation of beer bottles equals approximately 220M bottles.

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I have an odd hobby (odd because this interviewer was male) of knitting. I knit about 10 sweaters per year. I am looking into this as a business opportunity and want you to estimate the size of the hand knitting yarn market.

First I would look at all the places that sell hand-knitted sweaters and …

- *Actually, most of the hand knit sweaters sold in stores are produced abroad and the yarn they use is produced abroad. I am more concerned with the high quality hand knitting yarn sold in the U.S. for "hobby" type knitting.*

OK, then I would try to estimate how many people knit or I could look at how many stores sell knitting yarn.

- *Good, there are 3,000 specialty stores that sell knitting yarn. Also some bigger stores, like Wal-Mart, sell a small selection of lower quality yarn.*

OK, then I would take a sample of these stores and estimate their sales of yarn and then extrapolate that over the remaining stores. I would try to sample stores that are of typical size and revenues.

- *OK, I have data on 3 stores. The first is in Rhode Island and has $100,000 in annual revenues. The second store is in Austin Texas in the owner's garage. Its revenues are $40,000. The third store is in Massachusetts and has revenues of $170,000.*

I don't know if these three are typical and is not a large enough sample to base the system on. Hmmmmm . . . . . I guess if each store stays in business it must be making money. Maybe I can look at what it would take for each store to stay in business.

- *Good, what expenses would a store like this have?*

Rent, Labor, Advertising, and the cost of the products.

- *The mark-up on knitting supplies is about 100%, although with sale items it averages around 60%. Of the expenses, the variable costs make up about 50% of expenses. What would be your estimates for an average size store's expenses?*

Since these knitting shops are probably not in malls, and one was even in a garage, I would say the rent is fairly low, say $500 per month. There are probably just a few workers that aren't too well paid. Maybe they are making $30,000 a year.

- *Do you know what minimum wage is?*

OK, so maybe they make $15,000 a year. Advertising would consist of just local ads in papers and maybe a knitting magazine. I would guess that to be about $2,000 a year. The sum of these

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costs is $38,000 per year. If this is 50% of expenses, then the total costs would be $76,000. So with their 60% markup it would be….

- *For simplicity just base your initial estimates on break-even.*

OK, then 3,000 stores that sell $76,000 annually, it comes to $228 million. Wow that is bigger than I thought it would be.

- *Actually the knitting yarn market is about $350 million. Do you know why your numbers are understated?*

Well, actually I would say that my number is somewhat overstated because there are other supplies, such as knitting needles, patterns, etc that would be part of the sales. But I also realize it is understated because we assumed break-even and I am sure most of these stores turn a profit or they would not stay in business. Also, we only counted the sales at the specialty stores. I ignored the sales at the Wal-Mart type stores.

- *Good.*

How many people fly in and out of LaGuardia every day?

My first attempt at this was to begin with the number of airlines that fly into/ out of LGA. I then proceeded to try to figure out how many cities these airlines fly to from LGA. This was nearly impossible to determine realistically. I asked to try again and the interviewer said "good idea."

I looked at the problem again and realized it was a capacity problem. No two planes could be on a runway at a given time and most likely had to be spaced by a few minutes for safety reasons. With this assumption, I continued to break the day into peak (7am-10am, 3pm-8pm), midpeak (10am-3pm) and off peak times (8pm-11pm). I assumed no flights in the middle of the night. I further assumed planes are spaced 5 minutes apart at peak hours, 10 minutes at midpeak and 15 minutes apart during off peak times.

Capacity assumptions assumed 100% at peak, 75% at mid peak, and 50% at off peak. With an average plane holding 200 people, it would be (200 people/plane x 12 planes/hr x 8hrs) + (150 x 6 x 5) + (100 x 4 x 3) = 24,900 people.

With 2 runways, LaGuardia has roughly 50,000 people flying in and out every day.

The feedback I got was good (moved onto the next round of interviews). The interviewer told me he was looking for me to break the problem into peak and off peak times. To be even better you could give an answer for weekdays and weekends (peak times shift).

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How many gallons of ice cream are sold in the U.S. each year?

Ice cream can be sold through retailers and restaurants. First, let's analyze the retail sales. Assume that of 250 million people in the US, 80% like to eat ice cream. That makes 200 million possible consumers. Ice cream sales are likely to be somewhat seasonal especially in northern states, so assume an average selling season of eight months in the North and ten months in the South, for an average of nine months for the whole country. During the season, assume that people eat ice cream twice a month, and assume that the average serving is one pint. Since there are eight pints in a gallon, retail sales will be: 200 million people x 9 months x 2 servings per month x 1 pint / 8 pints per gallon = 450 million gallons.

Assume that 80% of the U.S. population frequents restaurants, and that they do so at a rate of twice per month on average. That makes 250 million people x 80% x 12 months per year x 2 visits per month = 4,800 million restaurant visits per month. Assume that 50% of these restaurants offer ice cream. That makes 4,800 million x 50% = 2,400 million possible purchases. Now assume that one out of ten times, the customer will order ice cream. That adds up to 2,400 million x 10% = 240 million purchases. Now assume that the average serving is half a pint. Since there are 16 half pints in a gallon, the total restaurant purchases come out to be 240 million purchases / 16 servings per gallon = 15 million gallons.

Total purchases of ice cream are 465 million gallons per year. Do a quick sanity check by dividing this number by 250 million people, which means that the average annual frozen yogurt consumption is 465/250 or a little less than 2 gallons per head of the population – that seems to be reasonable.

Are there two dogs in the world with the same number of hairs?

After a one-minute silence, the interviewer suggested that I divide the problem in 2 parts:

1. How many different possibilities are there for the number of hairs in a dog?
2. How many dogs are there in the world?

To find out the number of different possibilities of hair in a dog, I started by figuring out the hair-covered area of the smallest dog in the world and the largest dog in the world.

Approximate the body of a dog using geometrical figures: 1 cylinder for the body, four cylinders for the leg, 1 cylinder for the tail, 1 cylinder for the neck and 1 rectangular prism for the head.

For simplicity, the interviewer suggested that I used only the body area to calculate the number of hairs.

The area around a cylinder equals: π x diameter x length.

Each of the cylinders' two lids has an area of: π x radius^2.

Therefore, the total area of the cylinder equals:

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π x diameter x length + 2 x π x (diameter/2)^2.

I assumed the smallest dog in the world to be a newborn Chihuahua with a length of 10 cm and a diameter of 3 cm. Thus, the area of the newborn Chihuahua is (the interviewer allowed me to use π=3) 103.5 cm^2.

For the largest dog in the world I used an adult Saint Bernard with a length of 150 cm and a diameter of 50 cm. The area in the Saint Bernard's body is, therefore, 26,250 cm^2.

Then I ran into the problem of estimating the number of hairs in a square centimeter of dog skin. The interviewer suggested that I use 100 hairs. I asked the interviewer whether I could assume that all dogs, regardless of age and race, have the same hair density. He encouraged that to keep the problem simple.

So, according to our assumptions, the newborn Chihuahua has a total of 10,350 hairs while the adult San Bernardo has 2.625 million hairs.

Therefore, there are 2.625 million – 10,350 possibilities for the number of hairs in a dog, which I approximated to 2.61 million possibilities.

Now it is time to find out the number of dogs in the world. I let the interviewer know that I would exclude stray or organization owned (for security, etc) dogs from my analysis because I believe that most dogs live in households. He let me go ahead with my assumption.

I estimated the world population at 6 billion people. I assumed the average household size to be 5 people. Thus, there are 1.2 billion households in the world. I assumed that the percentage of households with dogs in the world was 30% and that the average number of dogs per household with dogs was 2. Therefore, my calculation for the number of dogs in the world is 720 million.

I now had the answers to the two parts in which the interviewer suggested that I divide the question but did not know what to do with them. I asked the interviewer whether I could assume if the possibilities of hairs in a dog were evenly distributed. The interviewer suggested that the probability was the same for any number of hairs.

I therefore assumed that the number of dogs for each possibility of number of hairs was equal and divided the total number of dogs by the total number of hair possibilities.

The result (720 million dogs in the world / 2.61 million hair possibilities) is 275 dogs per hair number possibility. Therefore, I concluded that YES, there are two dogs with the same number of hair in the world. But what if the question was "are there <u>exactly</u> two?"

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How many people have you interacted with over the last year?

Additional information provided (if interviewee asks probing questions):

- *Only count each unique person once (the interviewer complained that HBS students neglected this detail and came up with ridiculous answers like 13,000)*

One Approach:

Break into manageable subcategories and estimate them separately.

- CBS - almost 2,000 students, faculty and admin., assume I interact with 25%, so say 500.
- Social Settings - Events occur once or twice per week, more around the holidays, so say 100 events per year. The average number of people is on the order of 10 per event. Same people at different events, assume I see the average person 4 times. 100 events * 10 people / 4 times = 250 people. Maybe 50 of these people are also at CBS, so round down to 200 people.
- Everyday activities - dry cleaner, supermarket, favorite pizza place, post office, etc. I typically interact with a cashier and server, so assume 2 interactions per visit. Assume 3 errands or visits per day = 20 locations per week, average visit interval is once every two weeks, so there are 40 unique locations * 2 interactions = 80 people. Round up to 100 to account for my neighbors, doorman, my doctor, dentist, and other people I see over and over.
- Random meetings - people who stop you to ask for directions, people you talk to on the subway and people who attempt to steal your laptop or wallet - assume 2 people per week or 100 per yr.
- Other meetings - people you meet on vacation, at sporting events, shows, etc. Assume 50 people.

Total number of people in a year = 500 + 200 + 100 + 100 + 50 = approximately 1,000

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How big is the U.S. market for Band-Aids? (the brand)

- Band-Aids are used to cover up minor cuts. Assume that Band-Aid holds 75% of the U.S. market for bandages. The market can be segmented into two main categories of users: kids 16 and under who tend to get cuts more often, and adults over 16 who are more careful.
- Assume that the average life of a person is 80 years, and the population is evenly distributed. That means that kids 16 and under represent 16/80 = 20% of the population.
- Assume that they get a cut once every two months on average. If the U.S. population is 250 million, 20% equals 50 million kids. Once every two months equals six times per year, for a total of 50 million x 6 cuts = 300 million bandages.
- Assume that it takes three days on average to cure a cut and bandages are replaced once a day. That makes for 900 million bandages.
- The adults represent 80% of the 250 million people in the country, or 200 million.
- Assume that they get a cut once every six months that lasts three days, with bandages being replaced every day. That is 2 cuts per year x 3 days per cut x 200 million people = 1,200 million bandages.
- The total number of bandages, then, is 900 + 1,200 = 2,100 million bandages.
- Assume there are approximately 20 bandages in a package, and a package sells for $2. The total size of the market expressed in dollars is therefore 2,100 million / 20 x $2 which is approximately $200 million.
- Band-Aid holds 75% of this market which is equal to $150 million

How many pairs of skis do you expect to sell in the U.S. market as an up-market new entrant?

- Assume 250 million people in the U.S. 10% of those people ski which equals 25 million people.
- Assume a pair of skis lasts five years on average. This means that every year 1/5th of the skiing population buys a new pair of skis. That is 5 million pairs of skis per year.
- Now assume that 10% of the skiing population belongs to the "up-market" segment. Also assume that given the fanaticism and riches of this market segment, they replace their skis twice as often as the average person. That means that the market segment is 5 million x 10% x 2 = I million skis.
- Assume there are five major manufacturers in this segment at this time. That means that each sells 200,000 pairs of skis each year.
- Assume that as a new entrant, you will be able to attain 10% of the average sales volume in the first year. That is 10% x 200,000 pairs of skis = 20,000 pairs of skis.

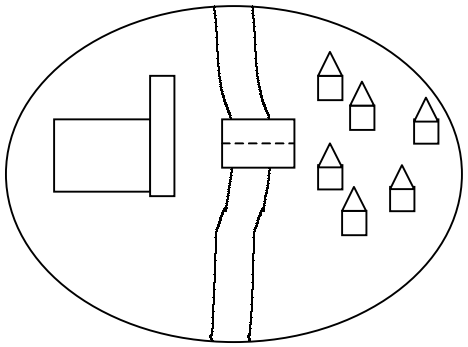
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Game Show and Creativity Cases

Picture a small town with a population of 10,000 people in rural Austria. A river divides the town. There is only one bridge with two lanes over this river and this is the only crossing point for several hundreds of kilometers. A factory stands on one side of the river and the entire population lives on the other side of the river.

The mayor of the village approaches you and tells you that the bridge presents a bottleneck to the village during rush hour when people are going to work (i.e. there are severe traffic jams). He wants you to solve the problem without spending a lot of money.

(The first thing I did was to draw a picture of the village.)



Just to make sure I understand this correctly, all villagers live on the right-hand side of the river, only the factory stands on the left-hand side of the river and only the villagers from the right-hand side work in this factory.

- Correct.

Is it possible to build another bridge?

- No. We want to keep this as cheap as possible.

Well, let's start out by identifying when these traffic jams occur. What time do the villagers go to work?

- There are two shifts; the first begins at 8am and the second starts at 9am. Each worker works 8 hours. Then they go home.

Okay. That implies that the traffic jams occur roughly between 7-9am, and 4-6pm. Do all men and women work at the factory?

- No. Only the men.

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Assuming 20% of the population is children, we are left with 8,000 adults. Assuming that the gender split in a typical Austrian village is 45% men - 55% women, we are left with 3,600 men that commute over the bridge. Does anyone besides the men have a reason to cross the bridge?

- No.

This means that, given a constant travel rate amongst all the men, roughly 1,800 men pass over the bridge in one hour (3,600 men / 2 hours). This translates into 30 men per minute (1,800 men / 60 minutes) and if each drives one car, 30 cars per minute.

- Okay. So now that you know when and why the traffic jams occur, what suggestions do you have to solve this dilemma? And please be as creative as possible.

(I could tell that I was on the right track. This guy was mainly looking for how creative I could be.) Well, given that most men travel over and back at approximately the same time, the mayor could give incentives to those men that car-pooled. The city could build car-pooling meeting points. This would eliminate a lot of traffic on the bridge. For example, if 3 men car-pooled every day, only 600 cars as opposed to 1,800 would pass over the bridge per hour (10 cars versus 30 per minute).

- Good.

A second suggestion would be to open both lanes to traffic. Between 7-9am, all traffic traveling west to the factory would be allowed to use both lanes. The opposite would apply to the afternoon rush hour period from 4-6pm.

- Good.

A third suggestion would be to subsidize those commuters that walked, used motor scooters or bicycles to get to work.

- Good.

(I could tell that this guy still wasn't all too impressed. I sat there and thought for a moment about my personal life and what experiences I had witnessed. Then it hit me.) My final suggestion resembles something I saw in Santiago de Chile when I lived there. The city had a serious problem with smog and as a result restricted the use of motor vehicles on certain days. But instead of restricting everyone's use, the city gave motorists different colored license plates that could only be used on a specific day of the week. So for example, if your car had a red license plate, you could only drive the vehicle on Mondays, Wednesday and Fridays. If you had a green license plate, you could only drive on Tuesdays, Thursday and Saturdays.

- Excellent! I have never heard that answer before. Good job.

After the last comment, I left the interview with a good feeling. I was lucky that I realized early on that the interviewer was more interested in the creative solutions I could come up with rather than just generic ones.

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You have two jars of wine: one of red wine and the other one of white wine. Each one is 100% pure. Now you take a glass and fill it with white wine and put it in the jar of red wine. You mix it and wait a couple of minutes. Then you take the glass and fill it in the jar which had originally 100% of red wine and put it in the jar with white wine. After doing this which jar is more pure with the original wine? (This is, is the jar which had originally 100% of red wine more "contaminated" with white wine or the other way?)

Intuitively the first thing that comes to my mind is that the white wine will be less contaminated with red wine since when you put back the glass of wine it will bring red and white wine…but on the other hand I also have to consider that what's not going back in the glass is staying…Can I change take a couple of minutes to think it?

- *Of course*

I have it now, the answer is the same.

- *That's correct, but how did you get it?*

Ok, there are two ways. First, assume hypothetically that when you take the glass back you are so lucky that you get all the white wine back. Therefore, all the white wine that contaminated the red wine is back on the white wine and we have the scenario of the beginning when they were both the same (100%). Now, lets think that we are less lucky and don't get a little part (lets say 1%) of the white wine. The glass will have 99% of white wine and 1% of red wine. Therefore, 1% of white wine stayed in the jar of red wine and 1% of red wine will end in the jar of white wine. So the final scenario is with both 99% pure. So, it will always be the same.

- *Very good, what's the other way you were thinking?*

Say it's not wine and its ten small balls, 10 red and 10 white. You take 3 white balls and put them with the red ones. Now you pick 3 balls from the 13 (10 red and 3 white) and it happens to be 2 red and 1 white. Both will finish with 10 balls: one will have 9 red and 1 white and the other one will have 9 white and 1 red.

- *Very good, lets go to another case.*

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What is 78 times 82? (no paper permitted)

It's about 80 times 80, which is 6,400.

More precise answer [what he was looking for]: It's a binomial so you can solve it as (80+2)*(80-2) which simplifies to 80-squared minus 2-squared or 6,396.

How many handshakes will eight people have to exchange when they are leaving the room?

The first person will have to shake seven hands, the second person will shake six hands, the third will shake five hands...etc.

7 + 6 + 5 + 4 + 3 + 2 + 1 = 28

You have a board of 64 squares of equal size (8 squares by 8 squares). You eliminate two of the board's corners that are diagonally opposed to one another. You are given a limitless number of dominos, which are each composed of two squares (same size as those of the board). Can you fill the board with dominos so that each remaining square is covered? (you may not juxtapose dominos)

No. Think of the board as a chessboard. Think of each domino as a rectangle of one black square and one white square. If you eliminate two diagonally opposed corners of a chess board, these corners will be of the same color (either both will be white or both will be black). Since you are eliminating two squares of the same color, you are eliminating two halves of two dominos instead of eliminating two squares of different color that could have been covered by one domino.

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There are two rooms: The Switch Room and the Bulb Room. In the Switch Room are three switches (1, 2, 3), all the switches are marked On and Off, and are originally in Off position. In the Bulb Room there are three light bulbs (A, B, C). You have to match the switches that turn on each light bulb. To do it, you start first in the Switch Room and do what ever you want with the three switches. Then you step into the Bulb Room and without going back to the Switch Room you have to figure out which switch controls which light bulb.

The problem is that there are three unknowns and only two equations. You need extra input. The solution is to turn on two lights, and turn one of them off after some time (let's say three minutes), and leave the third switch off, then you walk into the Bulb Room. There will be one light on which is controlled by the switch you left on. There will be two light bulbs off, but one of them will be warm (the extra input!), which is controlled by the switch left on for three minutes. The last light bulb, which will be cold, is controlled by the switch that you always left off.

You are in a room with two identical closed boxes. The boxes have identical tags that read "NPV of the contents of this box is one million dollars." What questions would you like to ask before you select one?

- What is the discount rate of each calculation?
- What is the time frame (and period length) of the two calculations?
- What is the range of possible outcomes of the two packages?
- What is the liquidity of each of the two packages?

Why are manholes round?

- So that the covers can't fall in the hole under any circumstances.
- So that the covers can be moved by rolling and no lifting is required.
- To provide the greatest opening area for the cost- and weight-limited material used.

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Why are soft drink beverage cans cylindrical?

- So that consumers won't cut their hands on the sharp edges. The shape is more comfortable and ergonomical. To maximize the ratio of the container's volume to its surface area short of using a sphere. This delivers more liquid per ounce of aluminum (thus, more per $).
- A sphere would be impractical because it would not stack or stand up after it was opened. Spheres would also require more air space between cans if they were in a box, vending machine or truck - a fact that could increase shipping and packaging costs.
- So that they will roll predictably and in control on assembly lines and vending machines.
- Because this container shape requires the least machining, joining, and finishing steps in manufacturing and is therefore the least expensive to manufacture.
- Circular structures distribute internal pressure. Further, structures with comers could develop fractures due to high stress at the edges.

Tell me all ways, practical or not, which you could use to determine whether a light goes off in the refrigerator when you close the door?

- With the door open, press the button that makes the light go on and off.
- Drill a hole in the door so that you can see inside when the door is closed.
- Find out the mean time to failure for these bulbs, close the door, and open it after the expiration time to see if the light is burned out.
- Go to the production line and perform a statistically valid test (appropriate number of samples) to determine whether the light always goes off (by pressing the button, etc.).
- Hook up and extremely sensitive electrical measuring device to the power source to see if the energy level drops when the door closes.
- Hook wires to the socket and perform a similar test when the door is closed.
- Place a sensitive thermometer (chilled to the refrigerator's temperature before testing) near the light bulb and close the door.
- Place some light sensitive material in the refrigerator to see if it is activated.
- Pick-up the phone, dial the manufacturer and ask if the light goes off when you close the door.
- "If no one is in there to see the light go off, does it matter?"

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Useful Facts & Conversions

You do not need to memorize all of these, but you should know the more basic ones such as the population of the United States (or wherever you will be applying for jobs) and how many feet in a mile.

POPULATIONS (2000)

- World 6.2 Billion
- Europe 730 Million
- Asia 3,700 Million
- United States 285 Million
- Canada 31 Million
- China 1,300 Million
- Select U.S. Cities:
  - New York City 8 Million
  - Los Angeles 3.8 Million
  - Chicago 2.9 Million

MEASUREMENTS

- **Distances**
  2.54 cm = 1 Inch
  12 inches = 1 Foot
  3 Feet = 1 Yard
  1 Mile = 5280 Feet = 1.61 Kilometers
- **Volume/Weight**
  1 cup = 8 ounces
  2 cups = 16 ounces = 1 pint (or 1 pound)
  4 cups = 2 pints = 1 quart
  4 quarts = 1 gallon
  2,000 pounds = 1 ton

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- **Height/Depth**
  Sears Tower, built 1974 = 1,454 ft
  Empire State, built 1931 = 1,250 ft
  Mt. Everest = 29,028 feet.
  Greatest known depth = Pacific Ocean, Mariana Trench = 35,810 feet
  Ocean depth at deepest point is about 7 miles (remember 5,280 ft/mile)
- **Length**
  Earth's diameter = 8,000 miles
  The Nile is the longest river in the world at 4,145 miles.
  The Great Wall of China stretches over 1,400 miles and can be seen from the Moon.

FACTS AND FIGURES

World's busiest airport - Passengers

1.	ATLANTA	- 80 Million
2.	CHICAGO	- 72 Million
3.	LOS ANGELES	- 68 Million
4.	LONDON	- 64 Million
5.	DALLAS/FT WORTH	- 60 Million
20.	NEW YORK	-32 Million

Wall Street Journal average daily circulation = 1,795,448
USA Today = 1,418,477
NY Times = 1,110,562

Worldwide auto production = 48 Million
U.S. auto production = 20% = 10 Million
Japanese auto production = 14 Million
Europe auto production = 18 Million

75% of Earth is covered in water
97% of that water is salt water

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CASE 37: CIGAR BAR

Category: Valuation

Question (posed by interviewer):

I was sitting in one of Chicago's new specialty "Cigar Bars" around the end of August with a friend. It was a Saturday night and the weather was fair. While enjoying one of the bar's finest stogies and sipping a cognac, I asked my friend how much he thought the bar was worth.

How would you go about determining the value of this bar?

Information to be given if asked:

Customers

- We arrived at the bar around 8:30pm. There appeared to be 30 customers already there. By 11pm the place had at least 70 customers. I would estimate the maximum capacity to be close to 100.

Products

- The bar sells two things: liquor and cigars.

Price

- The average cost of a cigar is $8 and the average cost of a drink is $7.

Employees

- There was one bar tender, a waiter and a waitress. All three were there the entire evening.

Miscellaneous

- The bar is located on one of Chicago's trendier streets with a lot of foot traffic.
- The bar is open Tuesday thru Sunday from 5 pm until 2 am.

Possible Solution:

This is a straightforward valuation. To perform a valuation you must estimate the cash flows from the business and discount them back using an appropriate weighted average cost of capital (WACC).

Revenues: One way to project revenues is to estimate the number of customers per day or per week and multiply that by the average expenditure of each customer. Keep in mind that Friday's and Saturday's are typically busier than other days and that people tend to be out more during the summer than in the winter.

Costs: There are two components to costs: fixed costs and variable costs. Under fixed costs you might consider: rent, general maintenance, management, insurance, liquor license, and possibly employees. The only real variable cost is the cost of goods sold.

Valuation: Subtract the costs from the revenues and adjust for taxes. You now have the annual cash flows generated from the bar. How long do you anticipate this bar being around? Cigar bars are a trend. In any case pick some number for the expected life (4-5 years). The discount rate should be a rate representative of WACC's of similar businesses with the same risk. Perhaps 20%. This gives you a value of:

$$\text{Value} = CF_1/1.2 + CF_2/(1.2)^2 + \ldots + Cf_n/(1.2)^n$$

CASE 38: NEW MAGAZINE

Category: Sizing

Question (posed by interviewer):

Your client is the CEO of a publishing company that produces a line of educational magazines as well as a line of women's magazines. Both businesses are profitable but are not growing quickly. He wants to start a third monthly magazine in the US targeted at 30-50 year old men (eg. GQ Magazine). His stated goal is to generate circulation revenues of $10 million in the first year. He has hired you to figure out whether this is possible.

Possible Solution:

This is an estimation case. The key here is to clearly define your assumptions, the specific answer is not important as long as you are making reasonable assumptions. For example

Target Customers
The total US population is approximately 240 million. Based on a normal distribution with the average life span of 80 years, approximately 2/3 of the population falls between 30-50 or about 160 million people. Approximately 1/2 are male or 80 million.

Of the 80 million 30-50 year old men in the country, assume that at least 1/2 would read a magazine or 40 million. Given the wide range of magazines on the market assume that only 10% of magazine readers would want to read a men's journal or 4 million target customers.

Share
As a new magazine assume that you can generate a 5% share of the men's magazine market in year one or 240,000 customers.

Revenues
Based on what other magazines sell for ($2.50-$5.00) assume a cover price. Lets say $3/magazine at the newsstand and $2/magazine for a subscription. Now make some assumptions on how many customers will buy on the newsstand versus subscription, lets say 50% subscribe (120,000) and 50% buy at the news stand (120,000). This comes out to $360,000 + $240,000 or $600,000. Finally, this is a monthly magazine. For simplicity assume that all target customers buy a magazine every month. This would generate total revenues of $600,000 X 12 or $7.2 million.

- In this case given the CEO's stated goal of $10 million in circulation revenues, it would not make sense to launch the magazine.

CASE 39: PIANO TUNERS

Category: Sizing

Question (posed by interviewer):

How many piano tuners are there in Chicago?

Approach

This is a brainteaser case. Its purpose is to test your logical and quick mathematical thinking. There is no right answer; the test is to see if you can come up with an answer based on the information you estimate.

You need to start by asking questions about the key factors. One way to solve it is to estimate the number of households in the Chicagoland area. The interviewer gave this piece of information at 2,000,000 households. Next, you can break the income of the households into four quarters (500,000 each). Make an estimate of 20% of highest income quarter have pianos, 10% of second quarter. 5% of third, and 0% of fourth.

Thus:

Income quarter	Population	% w/ Pianos	# of Pianos
1st	500,000	20%	100,000
2nd	500,000	10%	50,000
3rd	500,000	5%	25,000
4th	500,000	0%	0

With 175,000 pianos to tune you can estimate how often these pianos are tuned. You can estimate top income quarter tunes their pianos once a year, second quarter once every three years, third quarter once every 10 years. This gives you (100,000 + 50,000/3 + 25,000/10) = 119,167 or approximately 120,000.

Estimate a piano tuner can do four a day, 250 days a year, therefore: 120000/250=480 pianos a day to tune 480/4 = 120 pianos tuners needed.

How could you check this? Look in the yellow pages. Would all the piano turners be in there? You can guess half. By the way there are 46 piano tuners listed in the Chicago Yellow pages.

CASE 40: CHICAGO LOOP

Firm: Booz.Allen & Hamilton (2nd round – summer internship – 2001)
Type of Case: Market Sizing

Question (posed by the interviewer):
How would you go about estimating the daily average number of motor vehicles in the Chicago Loop?

Solution:

Author's Comments: Since I was not sure whether the interviewer wanted me to actually calculate my best estimate or just brainstorm creative ways to do the calculation, I decided to ask him beforehand. He offered me the following tradeoff: I could choose between going through the calculation or outlying alternative ways to arrive at an estimate. However, by choosing the latter, I had to come up with at least three different procedures (and that's exactly what I decided to do).

- 1st Approach: Secondary research. Contact the transit authority and see if the measurement has already been performed before. Similarly, downtown areas of cities of equivalent size could also have done some previous studies on the topic.
- 2nd Approach: Define the limits of the Loop and treat it as a system in which the units processed are motor vehicles. By measuring the rate in which cars get into and leave the system during the day, we could arrive at an estimate of the average "inventory" built into the system throughout the day. A sample of streets could be monitored (probably the most generally used) and the results could be extrapolated to the whole area.
- 3rd Approach: Use a phone directory to estimate the number of offices in the Loop. From there, estimate the number of people commuting to work everyday, and then the number of cars (discount people that car-pool or that use public transportation). This number will be a percentage of the total number of vehicles in the area, since we have to take into account the vehicles that are just driving through the Loop and heading somewhere else.
- 4th Approach: Measure the quality of the air in the Loop in terms of the concentration of gases (from fuel combustion) throughout the day and contrast it with the expected average contribution per vehicle.

Each suggestion was complemented with a discussion on pros and cons, regarding accuracy, costs, feasibility, time and resources needed.

CASE 41: CHEWING GUM MARKET

Category: Sizing

Question (posed by interviewer):

How would you estimate the size of the annual U.S. chewing gum market? Check your answer for reasonableness.

A typical approach:
Estimate the number of people who chew gum: of the 300 million population, 15% are between the ages of 10 and 20, the heaviest users, for a total of 45 million. Estimate that these people chew two packs per week, for annual sales of 4,500 million packs. For the other users over age 20, (70% of the 300 million population, or 210 million) estimate a usage rate of one half pack per week, for a total of 5,250 packs per year. Total packs per year is 9,750.

To check for reasonableness, figure the dollar sales that these packs represent: at 25 cents per pack, annual sales would be $2.4 billion, a reasonable figure.

CASE 42: GOLFBALL MARKET ENTRY

Category: Sizing

Question (posed by interviewer):

You are visiting a client who sells golf balls in the United States. Having had no time to do background research, you sit on the plane wondering what is the annual market size for golf balls in the U.S. and what factors drive demand. Your plane lands in fifteen minutes. How do you go about answering these questions?

Typical solution:

Golf ball sales are driven by end-users. The number of end users: take the population of 300 million; assume that people between 20 and 70 play golf (about 2/3 of the population, or 200 million) and estimate what proportion of these people ever learn to play golf (guess 1/4) which reduces the pool to 50 million. Now, estimate the frequency of purchase. If the average golfer plays twenty times per year, and requires two balls per time, that's forty balls per person. Multiply that times the 50 million, resulting in a 2 billion ball market.